

JAVA PROGRAM REPORT

Hybrid Inheritance Using Interfaces

Programming Language: Java

Topic: Hybrid Inheritance

Program File: HybridInheritance.java

Objective

To demonstrate hybrid inheritance in Java using interfaces, where interfaces B and C inherit from interface A and class D implements both B and C.

Java Source Code

```
package core_java;

interface A {
    void msg1();
}

interface B extends A {
    void msg2();
}

interface C extends A {
    void msg3();
}

class D implements B, C {

    // From Interface A
    public void msg1() {
        System.out.println("Message 1");
    }

    // From Interface B
    public void msg2() {
        System.out.println("Message 2");
    }

    // From Interface C
    public void msg3() {
        System.out.println("Message 3");
    }

    // Local method in Class D
    void msg4() {
        System.out.println("Message 4");
    }
}

public class HybridInheritance {
    public static void main(String[] args) {
        D obj = new D();

        obj.msg1();
        obj.msg2();
        obj.msg3();
        obj.msg4();
    }
}
```

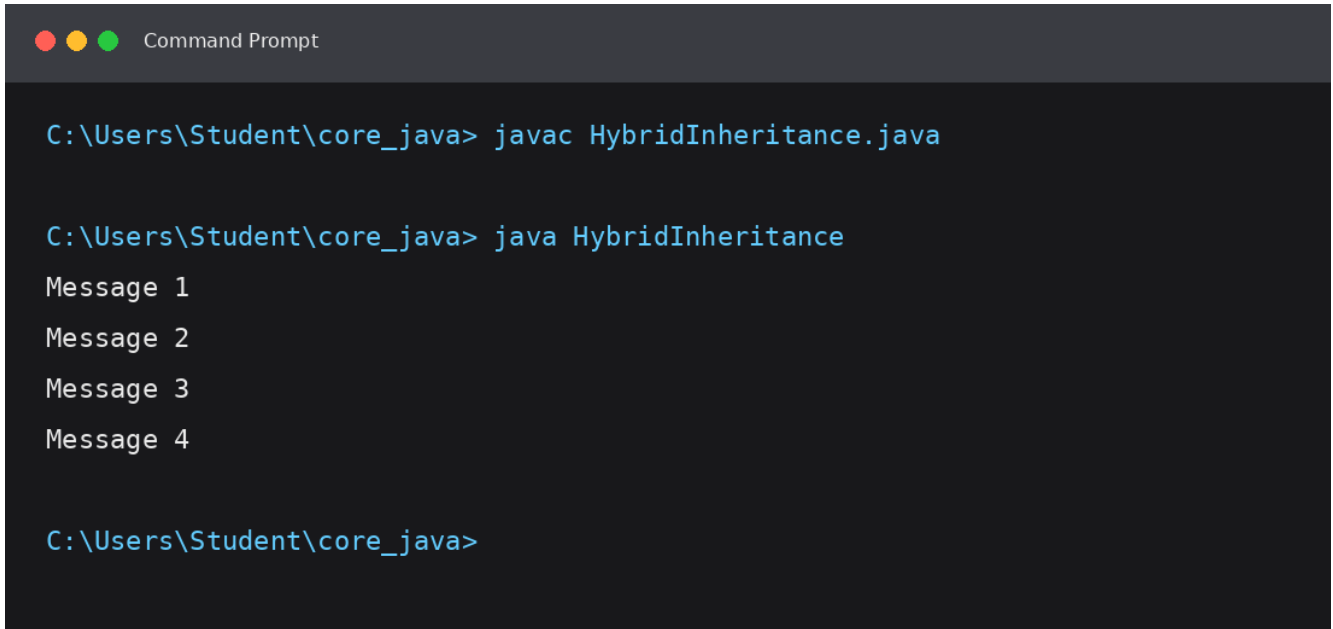
Program Explanation

- 1. Interface A:** This is the parent interface and declares the method `msg1()`.
- 2. Interface B:** Interface B extends A, so it inherits `msg1()` and additionally declares `msg2()`.
- 3. Interface C:** Interface C also extends A and additionally declares `msg3()`.
- 4. Class D:** Class D implements both B and C. Therefore, it provides implementations for `msg1()`, `msg2()`, and `msg3()`. It also contains its own local method `msg4()`.
- 5. Hybrid Inheritance:** The structure combines hierarchical and multiple inheritance through interfaces: A is extended by both B and C, while D implements both B and C.
- 6. Main Method:** An object of class D is created and all four methods are called sequentially.

Execution and Output Screenshots

The program was compiled and executed successfully. The compilation command is shown using the requested file name format: **javac HybridInheritance.java**.

Test Case 1: Compilation and Program Execution



```
Command Prompt

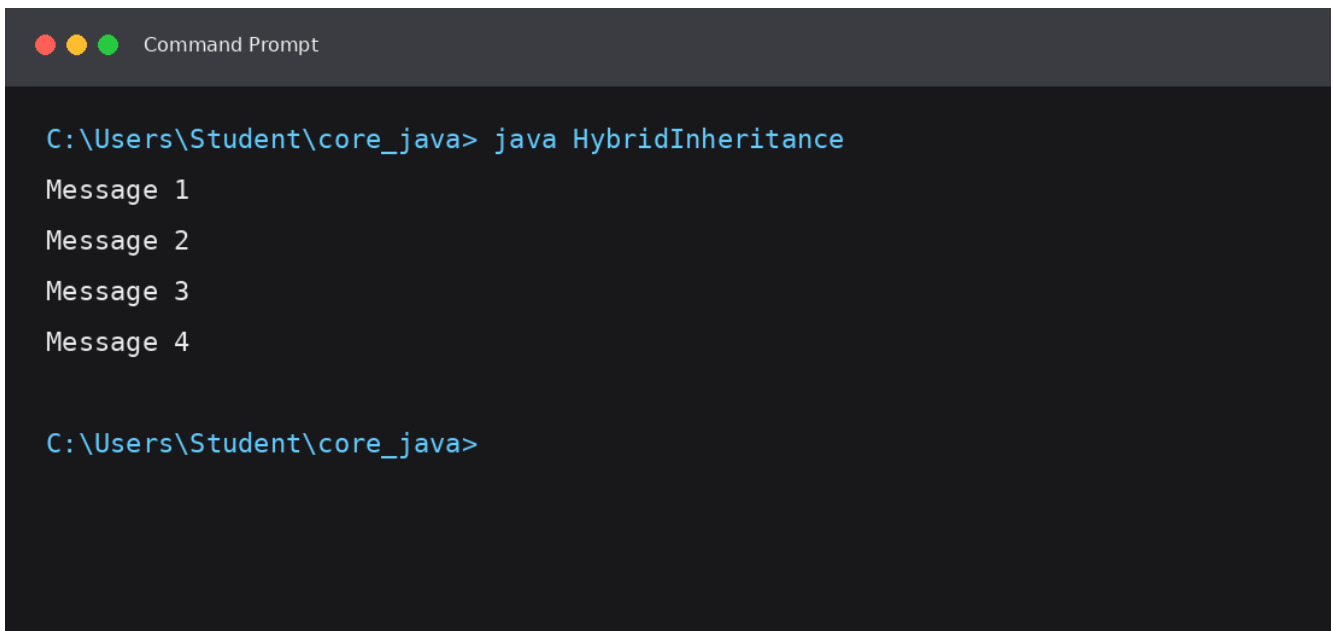
C:\Users\Student\core_java> javac HybridInheritance.java

C:\Users\Student\core_java> java HybridInheritance
Message 1
Message 2
Message 3
Message 4

C:\Users\Student\core_java>
```

Verification: The program compiled successfully using `javac HybridInheritance.java`. All four methods produced the expected messages.

Test Case 2: Re-execution Verification



```
Command Prompt

C:\Users\Student\core_java> java HybridInheritance
Message 1
Message 2
Message 3
Message 4

C:\Users\Student\core_java>
```

Verification: The program was executed again and produced the same output, verifying consistent behavior.

Test Case Summary

Test Case	Expected Result	Actual Result	Status
Test Case 1	Display Message 1 to Message 4	Output matched	PASS
Test Case 2	Same output on re-execution	Output matched	PASS

Conclusion

The program successfully demonstrates hybrid inheritance in Java through interfaces. Interfaces B and C inherit from A, and class D implements both interfaces. All required methods and the local method were executed successfully and the results were verified.